

Towards Optimality Preserving Aggregation for Scheduling Distributed Energy Resources

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Abstract—Scheduling the power exchange between a population of heterogeneous distributed energy resources and the corresponding upper level grid is an important control problem in power systems. A key challenge is the large number of (partially uncertain) variables that increase the computational burden and that complicate the structured consideration of uncertainties. Reducing the number of decision variables by means of aggregation can help alleviate these issues. However, despite the frequent use of aggregation for populations of storage devices, few works in the literature provide formal justification. In this article, we investigate aggregation of heterogeneous (storage) devices with time-varying power and energy constraints. In particular, we propose mild conditions on the constraints of each device, guaranteeing the applicability of an aggregated model in scheduling without any loss of optimality in comparison to the complete problem. We illustrate our findings on an relevant example.

Index Terms—Electric power networks, optimal control, optimization, problem reduction.

I. INTRODUCTION

THE ONGOING paradigm changes in the operation of power systems entail the need for coordinated management and operation of large populations of distributed energy resources (DERs) [1]. To this end, different explicit and implicit aggregation frameworks have been considered in the literature. This comprises concepts, such as *microgrids*, *virtual power plants*, and *integrated community energy systems*. Despite differences, all of the above consider dispatching active power across a (virtual/physical) boundary between some kind of lower level aggregated system and an upper level. This dispatch is often regulated by energy markets, and may need to be scheduled in advance [2]. Scheduling the power dispatch between a population of heterogeneous DERs (or an aggregation thereof) and the

corresponding upper level system is a pivotal problem that has received considerable research attention, cf., the overview [3] and the references therein. This problem is often tackled by means of optimization. The set of feasible schedules depends on the constraints on the directly controllable devices, and on the uncertain power outputs of loads and generators that are not directly controllable.

In this context, a model generating the feasible trajectories of the *aggregated* power output is preferable than a model predicting the feasible trajectories of the *individual* power outputs of each controllable device. The expected advantages of aggregated models are two-fold: computational scalability and improved handling of the uncertainties. Aggregated models are used in two different lines of research: 1) methods that compute the exact feasible set of the aggregated schedule and 2) methods entailing a relaxation of the feasible set of the aggregated schedule.

Examples of methods from group 1) are [4]–[7]. Therefore, the rule to disperse the aggregated state and/or input among the single devices has to be determined together with the aggregated feasible set. For example, Gorecki *et al.* [4] model the power output of each device to be an affine function of the aggregated schedule; Bernstein *et al.* [5], instead, considered ranges within which the states will be. However, these dispersion policies can be suboptimal. In contrast, Evans *et al.* [6], [7] considered a dispersion rule that they prove is always optimal. Nonetheless, the applicability of their method is restricted to devices with an unidirectional power output, i.e., it is limited to “discharging storage.”

Examples of methods from group 2), which entail a relaxation of the feasible set, are [8] for energy storage systems, [9]–[12] for plug-in electric vehicles, [13] and [14] for thermostatically controlled loads, and [15] and [16] for heterogeneous devices. The main idea of these methods is i) to describe each device as a “time-varying battery” [17]—i.e., storage with time-dependent power and energy constraints—and ii) to create an equivalent “time-varying battery” model for the aggregation by summing the bounds of the individual power and energy constraints. This method is simple, fully general, and does not require pre-determination of the individual responses of each device, thus allowing for greater flexibility. However, it may induce relaxed constraints, which can lead to solutions violating the original constraints of the individual devices. Consequently, applying such an aggregated model to scheduling calls guarantees the feasibility of the solution with respect to the original problem. The applicability and feasibility of aggregated models is largely investigated as a control problem, targeting aspects as frequency

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response [18] or system stability [19]. However, the majority of the works dealing with optimization-based control of power systems focuses on convex relaxations of the power flow equations (see [20]), neglecting reduced/aggregated models.

In this article, we investigate scheduling of an aggregation of heterogeneous energy-constrained DERs (EC-DERs) [6], such as storage or flexible loads, using a relaxed “time-varying battery” model of the aggregation. To the best of the authors’ knowledge, a thorough analysis of this aggregation concept has not been conducted yet. In particular, we derive sufficient conditions, on the constraint set and on the dispersion rule, that this kind of aggregated model can be used for scheduling without comprising feasibility of the individual constraints of the EC-DERs.

Our method provides an exact feasible set of the aggregated schedule by requiring only some conditions on the dispersion rule and on the set of constraints of each individual device. These conditions are such that they can be easily enforced by shrinking the original constraint set even when the model of the physical limitations of the system’s operational requirements does not meet the required conditions. This eventual preprocessing comes with multiple advantages if compared to other exact-model methods. First, the proposed conditions are a much less restrictive requirement than imposing a specific dispersion rule, e.g., as in [4]. Second, the proposed method can be applied for a larger class of problem than the one proposed by [6] and [7]. Put differently, it allows for time-varying energy constraints with bidirectional power outputs, while [6] and [7] are restricted to unidirectional power outputs, i.e., we consider both charging and discharging of the EC-DERs, while [6] and [7] consider discharging only. For this reason, not only do we provide theoretical results that support the works relying on a relaxed aggregated set—group 2), but, at the same time, we also go beyond existing works on exact aggregation—group 1). Last, as a byproduct of our investigation, we propose a methodology to ensure consistent online tracking of the aggregated schedule. This methodology is based on an appropriate dispersion (that we call *consistent dispersion*), computed via an optimization problem that can be solved pointwise in time.

The rest of this article is organized as follows: Section II presents the problem; Section III details a commonly used aggregated model and its limitations; Section IV describes the idea of a consistent dispersion of the aggregated stored energy, ensuring feasibility at the following step; Section V contains the main contribution of this article: A methodology to compute a consistent dispersion and the proof that such a dispersion always exists; Section VI outlines the discussion; Section VII summarizes our findings.

II. SCHEDULING FOR INTEGRATED ENERGY SYSTEMS

Subsequently, we introduce the problem analyzed throughout the article. First, we introduce the model of a single EC-DER. Then, we present a generic scheduling optimization problem. Finally, we propose a “time-varying battery” model as aggregation strategy and the corresponding aggregated scheduling.

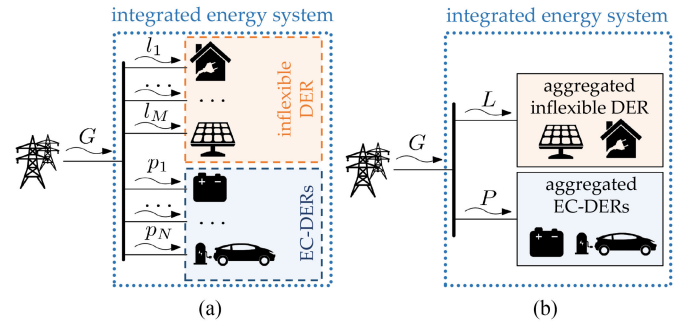


Fig. 1. Schematic representation of an integrated energy system.

A. Energy-Constrained Distributed Energy Resources

A large subset of the technologies, referred to as DER, is *energy-constrained* [6]. EC-DERs are subject to dynamics modeling either the amount of physically stored energy or the integral amount of energy exchanged between the EC-DER and an upper level entity over a given time period. Examples are batteries and other types of storage, as well as flexible demand.

In this article, we consider a population $\mathcal{N} = \{1, \dots, N\}$ of heterogeneous EC-DERs. Following [15],¹ we describe the j th EC-DER by

$$e_j(k+1) = e_j(k) + \delta \cdot p_j(k), \quad e_j(0) = e_j^0 \quad (1)$$

with time index $k \in \mathbb{N}$, sampling period δ , and initial condition e_j^0 . The variable $p_j(k) \in \mathbb{R}$ is the (averaged) power output over the k th step, and $e_j(k+1) \in \mathbb{R}$ is the energy state at time k . Both $p_j(k)$ and $e_j(k)$ are subject to time-varying constraints

$$p_j(k) \in \mathcal{P}_j(k) \quad (2a)$$

$$e_j(k+1) \in \mathcal{E}_j(k+1) \quad (2b)$$

with $\mathcal{P}_j(k)$ and $\mathcal{E}_j(k)$ being closed real intervals, i.e., $\mathcal{P}_j(k) = [p_j(k), \bar{p}_j(k)] \subset \mathbb{R}$ and $\mathcal{E}_j(k+1) = [e_j(k+1), \bar{e}_j(k+1)] \subset \mathbb{R}$.

The model comprised by (1–2) admits noteworthy properties: First, it neglects conversion losses and reactive power; we will comment on this in Section VI-B. Second, constraints (2a) and (2b) are time varying. Time-varying energy constraints allow to model charging requests (by varying the lower bound $\underline{e}_j$) and availability windows (by varying the upper bound $\bar{e}_j$). In turn, time-varying energy constraints require time-varying power constraints, as later clarified in Section V. Finally, observe that (2a) and (2b) are decoupled and consider the operation of each device independently from the others. Additional comments on this point will also follow in Section V.

B. Scheduling With Individual Device Models

Consider a microgrid including M inflexible DERs and loads, and N EC-DERs, as illustrated in Fig. 1(a). There, $G(k)$ indicates the aggregated (average) active power exchange between the internal and the external level at step k ; $l_i(k)$ is the aggregated (average) active power output over time step k of the

¹Note that the authors of [15] refer to EC-DERs as “flexible loads”.

i th inflexible DER/load with $i \in \mathcal{M} = [1, \dots, M]$; and $p_j(k)$ is the aggregated (average) active power output over time step k of the j th EC-DER with $j \in \mathcal{N}$. The connections are assumed to be lossless and such that the system components are able to exchange power mutually without any technical limit. This assumption is justified for groups of devices located within a limited geographical area, as shown in [21]. Let us introduce the vector notation

$$\begin{aligned} \mathbf{l}(k) &\doteq [l_1(k), \dots, l_M(k)] \in \mathbb{R}^M \\ \mathbf{p}(k) &\doteq [p_1(k), \dots, p_N(k)] \in \mathbb{R}^N \\ \mathbf{e}(k) &\doteq [e_1(k), \dots, e_N(k)] \in \mathbb{R}^N. \end{aligned}$$

Then, the power balance at time k reads

$$G(k) = \mathbf{1}^\top \mathbf{p}(k) - \mathbf{1}^\top \mathbf{l}(k)$$

with $\mathbf{1}$ being a vector of ones whose dimension follows from context.

Consider the problem of optimizing the power exchange $G(k)$ over a time horizon $\mathcal{K} = [0, \dots, K-1] \subset \mathbb{N}$ with respect to a given cost function $c: \mathbb{R} \rightarrow \mathbb{R}$. Assuming that the system constraints are given and the values of $l_i(k)$ over $\mathcal{K}$ are known for all $i \in \mathcal{M}$, the scheduling problem with individual dynamics reads

$$\min_{\substack{\{G(k)\}_{\mathcal{K}}, \\ \{\mathbf{p}(k)\}_{\mathcal{K}}, \{\mathbf{e}(k+1)\}_{\mathcal{K}}}} \sum_{k \in \mathcal{K}} c(G(k)) \quad (3a)$$

$$\text{s. t. } \forall k \in \mathcal{K}$$

$$G(k) = \mathbf{1}^\top \mathbf{p}(k) - \mathbf{1}^\top \mathbf{l}(k) \quad (3b)$$

$$\mathbf{e}(k+1) = \mathbf{e}(k) + \delta \cdot \mathbf{p}(k), \quad \mathbf{e}(0) = \mathbf{e}^0 \quad (3c)$$

$$\mathbf{p}(k) \in \mathcal{P}(k) \quad (3d)$$

$$\mathbf{e}(k+1) \in \mathcal{E}(k+1). \quad (3e)$$

Notation $\{\cdot\}_{\mathcal{K}}$ indicates the sequence of $\cdot$ over $k \in \mathcal{K}$.

The constraints are as follows. Equality (3b) is the power balance and (3c)–(3e) model the EC-DERs, see (2) and (1). The sets $\mathcal{P}(k)$ and $\mathcal{E}(k)$ are hyperboxes of $\mathbb{R}^N$ built upon the constraints of the single devices, i.e.,

$$\mathcal{P}(k) = \mathcal{P}_1(k) \times \dots \times \mathcal{P}_N(k) \subset \mathbb{R}^N \quad (4a)$$

$$\mathcal{E}(k) = \mathcal{E}_1(k) \times \dots \times \mathcal{E}_N(k) \subset \mathbb{R}^N. \quad (4b)$$

C. Scheduling and Aggregated Models

Upon solving (3)—and provided it is feasible— $\{\mathbf{p}(k)\}_{\mathcal{K}}$ and $\{\mathbf{e}(k+1)\}_{\mathcal{K}}$ are obtained. However, in terms of coordination with the upper level grid, the schedule itself, i.e., $\{G(k)\}_{\mathcal{K}}$, is the only information of crucial interest. Moreover, it is also the only decision variable entering the cost function directly. Thus, the question arises whether it is possible to reduce the number of constraints and decision variables in (3) by means of aggregation.

The first step in aggregating DERs is to cluster the power outputs of DERs in inflexible ones and energy-constrained ones,

see Fig. 1(b). To this end, consider the function

$$s: \mathbb{R}^{n_x} \rightarrow \mathbb{R}, \quad \mathbf{x} \mapsto X = \mathbf{1}^\top \mathbf{x} \quad (5)$$

which sums—i.e., *aggregates*—the elements of a vector $\mathbf{x} \in \mathbb{R}^{n_x}$. Formally, its inverse is the set-valued (preimage) map

$$s^{-1}: \mathbb{R} \rightarrow \mathbb{R}^{n_x}, \quad s^{-1}(X) = \{\mathbf{x} \in \mathbb{R}^{n_x} \mid X = \mathbf{1}^\top \mathbf{x}\}. \quad (6)$$

We refer to $s^{-1}(X) \subset \mathbb{R}_x^n$ as *the set of dispersions of $X \in \mathbb{R}$ to $\mathbf{x} \in \mathbb{R}^{n_x}$* .² The aggregated power outputs and the aggregated energy read

$$L(k) \doteq s(\mathbf{l}(k)), \quad P(k) \doteq s(\mathbf{p}(k)), \quad E(k) \doteq s(\mathbf{e}(k)). \quad (7)$$

These aggregated variables allow rewriting (3) as follows:

$$\min_{\substack{\{P(k)\}_{\mathcal{K}}, \{E(k+1)\}_{\mathcal{K}}, \\ \{\mathbf{p}(k)\}_{\mathcal{K}}, \{\mathbf{e}(k+1)\}_{\mathcal{K}}}} \sum_{k \in \mathcal{K}} c(P(k) - L(k)) \quad (8a)$$

$$\text{s. t. } \forall k \in \mathcal{K}$$

$$\mathbf{e}(k+1) = \mathbf{e}(k) + \delta \cdot \mathbf{p}(k), \quad \mathbf{e}(0) = \mathbf{e}^0 \quad (8b)$$

$$\mathbf{p}(k) \in \mathcal{P}(k) \quad (8c)$$

$$\mathbf{e}(k+1) \in \mathcal{E}(k+1) \quad (8d)$$

$$P(k) = s(\mathbf{p}(k)) \quad (8e)$$

$$E(k+1) = s(\mathbf{e}(k+1)) \quad (8f)$$

which is equivalent to (3) with the additional decision variables $\{P(k)\}_{\mathcal{K}}$ and $\{E(k)\}_{\mathcal{K}}$, and with decision variable $G(k)$ directly replaced with the summation of $P(k)$ and $L(k)$.

Aggregated scheduling builds upon (8), by dropping the constraints on $\mathbf{p}(k)$ and $\mathbf{e}(k)$ [i.e., (8b)–(8f)] and substituting them with constraints on the aggregated variables. This yields

$$\min_{\{P(k)\}_{\mathcal{K}}, \{E(k+1)\}_{\mathcal{K}}} \sum_{k \in \mathcal{K}} c(P(k) - L(k)) \quad (9a)$$

$$\text{s. t. } \forall k \in \mathcal{K}$$

$$E(k+1) = E(k) + \delta \cdot P(k), \quad E(0) = s(\mathbf{e}^0) \quad (9b)$$

$$P(k) \in \mathcal{P}_{\text{agg}}(k) \quad (9c)$$

$$E(k+1) \in \mathcal{E}_{\text{agg}}(k+1). \quad (9d)$$

The aggregated problem avoids computing $\{\mathbf{p}(k)\}_{\mathcal{K}}$ and $\{\mathbf{e}(k)\}_{\mathcal{K}}$ explicitly. Problem (8) and (9) have the same cost function. Thus, the solutions $\{P(k)\}_{\mathcal{K}}$ —and $\{E(k+1)\}_{\mathcal{K}}$ —of (8) and (9) coincide whenever (9b)–(9d) and (8b)–(8f) describe the same constraint set for $\{P(k)\}_{\mathcal{K}}$ and $\{E(k+1)\}_{\mathcal{K}}$.³ Using (9) instead of (3) for scheduling purposes requires an appropriate construction of the sets $\mathcal{P}_{\text{agg}}(k)$ and $\mathcal{E}_{\text{agg}}(k)$. However, the construction of such sets is ambiguous. In the following, we discuss a relaxation to compute $\mathcal{P}_{\text{agg}}(k)$ and $\mathcal{E}_{\text{agg}}(k+1)$ that is of practical relevance.

²This notion is chosen to the end of avoiding confusion with *statistical distributions* of random variables, which are also of interest in scheduling problems.

³Recall that dynamics (9b) implies that $\{E(k+1)\}_{\mathcal{K}}$ is uniquely determined by the sequence $\{P(k)\}_{\mathcal{K}}$.

III. CALCULATING BOUNDS FOR THE AGGREGATED MODEL

A common practice to construct the sets $\mathcal{P}_{\text{agg}}(k)$ and $\mathcal{E}_{\text{agg}}(k)$ (see for example [15]) is to consider intervals of $\mathbb{R}$ given by the (pointwise) mapping of the hyperboxes $\mathcal{P}(k)$ and $\mathcal{E}(k)$

$$\mathcal{P}_{\text{agg}}(k) = s(\mathcal{P}(k)), \quad \mathcal{E}_{\text{agg}}(k) = s(\mathcal{E}(k)) \quad (10)$$

where $s(\mathcal{X})$ denotes the pointwise application of s to $x \in \mathcal{X}$

$$s(\mathcal{X}) = \{X = s(\mathbf{x}) \mid \mathbf{x} \in \mathcal{X}\}.$$

Practically speaking, this choice leads to

$$\mathcal{P}_{\text{agg}}(k) = [\underline{P}(k), \bar{P}(k)], \quad \mathcal{E}_{\text{agg}}(k) = [\underline{E}(k), \bar{E}(k)] \quad (11a)$$

with

$$\underline{P}(k) = \sum_{j \in \mathcal{N}} \underline{p}_j(k), \quad \bar{P}(k) = \sum_{j \in \mathcal{N}} \bar{p}_j(k) \quad (11b)$$

and

$$\underline{E}(k) = \sum_{j \in \mathcal{N}} \underline{e}_j(k), \quad \bar{E}(k) = \sum_{j \in \mathcal{N}} \bar{e}_j(k). \quad (11c)$$

This way the aggregated sets $\mathcal{P}_{\text{agg}}(k)$ and $\mathcal{E}_{\text{agg}}(k)$ are computed independently at each time step, and do not require to predetermine any dispersion rule. This methodology is utilized in several works, cf., group 2) in the introduction. The underlying ratio stems from interval arithmetics [22]. Indeed, it can be shown that if $\mathbf{p}(k) \in \mathcal{P}(k)$, then

$$P(k) = s(\mathbf{p}(k)) \in s(\mathcal{P}(k)).$$

The same holds for the aggregated energy $E(k)$. Thus, feasibility of $P(k)$ and $E(k)$ in (8c)–(8f) implies their feasibility with respect to (9c)–(9d). Moreover, it is easy to see that if $\mathbf{e}(k)$ and $\mathbf{p}(k)$ satisfy (8b), then $P(k)$ and $E(k)$ satisfy (9b). To summarize, considering (10), the aggregated dynamics (9b), and the constraints (9c)–(9d) do not restrict the feasible set of (8). Rather, they are a relaxation of the propagation of the original constraint set in (3). However, this can cause situations in which the aggregated schedules of $P(k)$ and $E(k)$ cannot be dispersed without violating individual constraints on $\mathbf{p}(k)$ and $\mathbf{e}(k)$. Next, we illustrate this issue with a simple example, where the scheduling horizon is restricted to a single step in the future, $k+1$, see also paragraph preceding (3).

Example 1 (Nondispersable aggregation): Consider two EC-DERs, i.e., $\mathcal{N} = \{1, 2\}$, and two subsequent time instants, k and $k+1$. For the sake of simplicity, let the EC-DERs have identical time-invariant power and energy limits, i.e., $\mathcal{P}_1(k) = \mathcal{P}_2(k) = [\underline{p}, \bar{p}] \subset \mathbb{R}$, and $\mathcal{E}_1(k) = \mathcal{E}_2(k) = \mathcal{E}_1(k+1) = \mathcal{E}_2(k+1) = [\underline{e}, \bar{e}] \subset \mathbb{R}$. Furthermore, let $\bar{p} = -\underline{p}$ with $2\bar{p}\delta \leq \bar{e} - \underline{e}$. ■

Now, consider the case in which $e_1(k) = \bar{e}$ and $e_2(k) = \underline{e}$, implying $E(k) = \underline{e} + \bar{e}$. The constraints limiting $e_1(k+1)$ and $e_2(k+1)$ are depicted in Fig. 2(a) and (b), respectively. The blue striped interval represents the feasible states according to the energy constraint; the yellow cone indicates the states admissibly reachable from $e_1(k)$ and $e_2(k)$. According to (10), the aggregated constraints are

$$P(k) \in [2\underline{p}, 2\bar{p}], \quad E(k+1) \in [2\underline{e}, 2\bar{e}] \quad (12a)$$

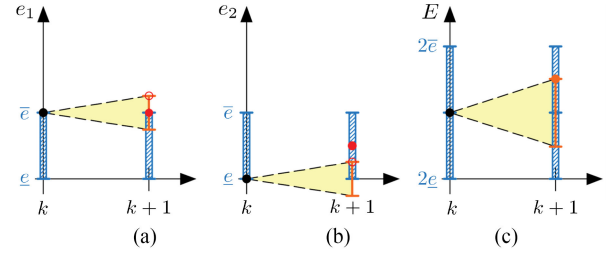


Fig. 2. Illustration of Example 1.

For the sake of illustration, consider $E(k+1) = \underline{e} + \bar{e} + \delta \cdot 2\bar{p}$; the corresponding value of $P(k) = 2\bar{p}$ follows. This choice is feasible for the aggregated model; it satisfies (12). In Fig. 2(c), the choice of $E(k+1)$ is depicted by an orange circle. Note that the dispersion of $E(k+1)$ and $P(k)$ to individual devices leads to infeasibility on the individual side.

However, a feasible choice of $\mathbf{p}(k)$, such that $\mathbf{p}(k) = s^{-1}(2\bar{p})$ exists, it is

$$p_1(k) = \bar{p} \in [\underline{p}, \bar{p}], \quad p_2(k) = \bar{p} \in [\underline{p}, \bar{p}].$$

Yet, this implies

$$e_1(k+1) = \bar{e} + \delta \cdot \bar{p} \notin [\underline{e}, \bar{e}], \quad e_2(k+1) = \underline{e} + \delta \cdot \bar{p} \in [\underline{e}, \bar{e}]$$

which violates the energy constraint of EC-DER $j = 1$. This can be seen in Fig. 2(a) and (b), where empty red circles indicate energy values, which sum up to $E(k+1)$ and are reachable from $e_1(0), e_2(0)$ but not feasible with respect to $e_1(k+1) \in [\underline{e}, \bar{e}]$.

At the same time, a feasible choice of $\mathbf{e}(k+1)$ such that $\mathbf{e}(k+1) = s^{-1}(\underline{e} + \bar{e} + \delta \cdot 2\bar{p})$ is

$$e_1(k+1) = \bar{e} \in [\underline{e}, \bar{e}], \quad e_2(k+1) = \underline{e} + \delta \cdot 2\bar{p} \in [\underline{e}, \bar{e}].$$

This implies $p_1(k) = \frac{\bar{e} - \underline{e}}{\delta} = 0 \in [\underline{p}, \bar{p}]$, and

$$p_2(k) = \frac{\underline{e} - (\underline{e} + \bar{e} + 2\bar{p})}{\delta} \notin [\underline{p}, \bar{p}]$$

which violates the power constraint of EC-DER $j = 2$. This is depicted in Fig. 2(a) and (b) by red circles, which indicate energy values summing up to $E(k+1)$ that are feasible but not reachable for EC-DER $j = 2$. Hence, the chosen values for $E(k+1)$ and $P(k)$ are feasible for the aggregated model, but cannot be dispersed to $\mathbf{p}(k)$ and $\mathbf{e}(k+1)$. ■

Example 1 illustrates that an aggregated model with power and energy constraint computed as in (10) might lead to infeasibilities in practice. The timewise coupling between the variables $p_j(k)$ and $e_j(k)$ —and, consequently, between constraints (2) introduced by the discrete-time dynamics (1) is the main source of these infeasibilities. In other words, aggregation as in (10) ignores the reachability properties of the dynamics of the single devices.

Hence, given the initial condition $e_j(k)$ and the system dynamics (1), we derive a one-step energy constraint collecting the power and the energy constraints (2) at $k+1$. Given $e_j(k)$, the power constraint (2a) can be seen as an implicit constraint on $e_j(k+1)$ because it limits the energy state that can be reached at the following step $k+1$. This can be formalized considering the one-step reachable set of (1)

$$e_j(k) \oplus \delta \cdot \mathcal{P}_j(k). \quad (13)$$

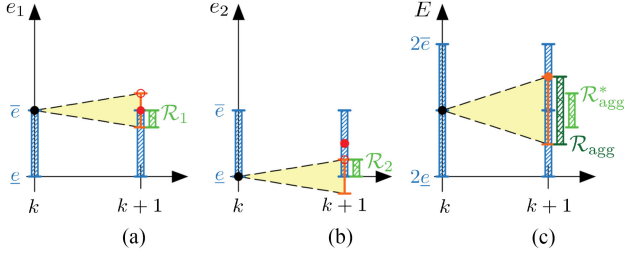


Fig. 3. Illustration of the energy constraints (14) and (18) in Example 1.

Here, $\oplus$ denotes the Minkowski sum. Note a slight abuse of notation: for the sake of readability, the set $\{e_j(k)\}$ is indicated with its sole element $e_j(k)$. However, not all $e_j(k+1) \in e_j(k) \oplus \delta \cdot \mathcal{P}_j(k)$ are feasible with respect to the energy constraint at $k+1$ (2b). Combining (2) and (1) leads to

$$e_j(k+1) \in \mathcal{R}_j(k+1, e_j(k)) \quad (14)$$

where

$$\mathcal{R}_j(k+1, e_j(k)) \doteq \mathcal{E}_j(k+1) \cap (e_j(k) \oplus \delta \cdot \mathcal{P}_j(k)). \quad (15)$$

The set $\mathcal{R}_j(k+1, e_j(k))$ is the interval of feasible and reachable $e_j(k+1)$ given $e_j(k)$. For the sake of readability, we omit the explicit dependency of the interval $\mathcal{R}_j(k+1, e_j(k))$ on $e_j(k)$ in the sequel. Evidently, one needs to avoid cases in which the intersection in (15) is the empty set. To this end, we consider the following assumption.

Assumption 1 (Controlled forward invariance): For all $j \in \mathcal{N}$ EC-DERs and all $k \in \mathcal{K}$, let the sampling time δ and the intervals $\mathcal{P}_j(k)$, $\mathcal{E}_j(k)$, and $\mathcal{E}_j(k+1)$ be such that

$$\mathcal{E}_j(k+1) \cap (e_j(k) \oplus \delta \cdot \mathcal{P}_j(k)) \neq \emptyset \quad (16)$$

holds for all $e_j(k) \in \mathcal{E}_j(k)$. ■

This condition can be regarded as controlled forward invariance of the entire (time-varying) state constraint set $\mathcal{E}_j(k)$, $k \in \mathcal{K}$. Under Assumption 1, the bounds of

$$\mathcal{R}_j(k+1) = [\underline{e}_j^r(k+1), \bar{e}_j^r(k+1)] \quad (17a)$$

$$\underline{e}_j^r(k+1) = \max \{ \underline{e}_j(k+1), e_j(k) + \delta \cdot \underline{p}_j(k) \} \quad (17b)$$

$$\bar{e}_j^r(k+1) = \min \{ \bar{e}_j(k+1), e_j(k) + \delta \cdot \bar{p}_j(k) \} \quad (17c)$$

which follows from standard tools of interval arithmetics [22]. Going back to Example 1, a sketch of $\mathcal{R}_1(k+1)$ and $\mathcal{R}_2(k+1)$ is shown in Fig. 3(a) and (b). Therefore, these intervals are depicted by green crossed intervals.

Similarly, a one-step aggregate energy constraint substituting the aggregated power and the energy constraints can be elaborated. This model requires a joint power and energy constraint (per time step), i.e.,

$$E(k+1) \in \mathcal{R}_{\text{agg}}(k+1, E(k)) \quad (18)$$

with

$$\mathcal{R}_{\text{agg}}(k+1, E(k)) \doteq \mathcal{E}_{\text{agg}}(k+1) \cap (E(k) \oplus \delta \cdot \mathcal{P}_{\text{agg}}(k)).$$

Henceforth, we simplify the notation by dropping the explicit dependency of $\mathcal{R}_{\text{agg}}(k+1, E(k))$ on $E(k)$. Similar to (17), the

bounds of

$$\mathcal{R}_{\text{agg}}(k+1) = [\underline{E}^r(k+1), \bar{E}^r(k+1)] \quad (19a)$$

$$\underline{E}^r(k+1) = \max \{ \underline{E}(k+1), E(k) + \delta \cdot \underline{P}(k) \} \quad (19b)$$

$$\bar{E}^r(k+1) = \min \{ \bar{E}(k+1), E(k) + \delta \cdot \bar{P}(k) \}. \quad (19c)$$

However, given $\mathbf{e}(k)$, the actual aggregation of the one-step feasible and reachable energy intervals of each EC-DER, $\mathcal{R}_j(k+1)$, gives

$$\mathcal{R}_{\text{agg}}^*(k+1) \doteq s(\mathcal{R}(k+1)) \quad (20)$$

where

$$\mathcal{R}(k+1) = \mathcal{R}_1(k+1) \times \dots \times \mathcal{R}_N(k+1) \subset \mathbb{R}^N$$

is a hyperbox of $\mathbb{R}^N$ built upon the combined constraints of the single devices (17). The propagation of set $\mathcal{R}(k+1)$ with s from (5) is again to be understood as the pointwise image. This can be written as

$$s(\mathcal{R}(k+1)) = \left[\sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1), \sum_{j \in \mathcal{N}} \bar{e}_j^r(k+1) \right].$$

Next, we investigate the relation between the bounds of $\mathcal{R}_{\text{agg}}(k+1)$ and $\mathcal{R}_{\text{agg}}^*(k+1)$. The application of (10) to (19) implies the following inequalities:

$$\underline{E}^r(k+1) \leq \sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1) \quad (21a)$$

$$\bar{E}^r(k+1) \geq \sum_{j \in \mathcal{N}} \bar{e}_j^r(k+1). \quad (21b)$$

Consequently, we have that

$$\mathcal{R}_{\text{agg}}(k+1) \supseteq \mathcal{R}_{\text{agg}}^*(k+1). \quad (22)$$

To summarize, the aggregated constraints from (10) may enlarge the feasible space of $P(k)$ and $E(k)$, even in the simple case of a one-step scheduling horizon. This is the case of Example 1, illustrated in Fig. 3(c). It can be seen that the interval $\mathcal{R}_{\text{agg}}^*(k+1)$ is a “narrower” subset of $\mathcal{R}_{\text{agg}}(k+1)$. The value of $E(k+1)$ chosen in Example 1 is feasible and reachable according to the aggregated model, as it is contained within $\mathcal{R}_{\text{agg}}(k+1)$. However, this solution is infeasible for the complete constraint set, because it is *outside* of $\mathcal{R}_{\text{agg}}^*(k+1)$. The key point is that aggregation as in (10) ignores the dispersion of the initial state, which may further limit the reachable set at the next step. Subsequently, we present sufficient conditions under which aggregated constraints as in (10) equal the propagation of the original constraints.

IV. ENSURING SCHEDULE FEASIBILITY

The analysis in the previous section implies that (10) cannot guarantee the equivalence of the two feasible sets, because it ignores the way in which $E(k)$ is dispersed. Moving to a multistep horizon, it seems that (9) cannot be used as a substitute for (8) if (10) defines the feasible set of (9). Indeed this observation motivates aggregation methods that account for the dispersion rule explicitly, cf., the references listed in group 1) in

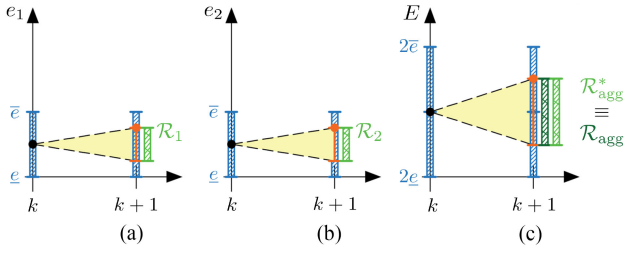


Fig. 4. Illustration of Example 2.

the introduction. In this article, we pursue a different approach: instead of computing $\mathcal{R}_{\text{agg}}(k+1)$ knowing the dispersion rule, we find *conditions* for the dispersion of $E(k)$ under which

$$\mathcal{R}_{\text{agg}}(k+1) \equiv \mathcal{R}_{\text{agg}}^*(k+1) \quad (23)$$

holds with $\mathcal{R}_{\text{agg}}(k+1)$ following from $\mathcal{P}_j(k)$ and $\mathcal{E}_j(k+1)$ computed via (10). We clarify this concept via the following example.

Example 2: Consider the setting of Example 1, but with $e_1(k) = e_2(k) = 0.5(\bar{e} + \underline{e})$. Fig. 4 depicts this case. Therefore, $\mathcal{R}_{\text{agg}}(k+1)$ and $\mathcal{R}_{\text{agg}}^*(k+1)$ are equivalent and $E(k+1) = 2\bar{e}$ —which is feasible and reachable for the aggregated model as discussed in Example 1 is feasible and reachable with respect to the constraints of each EC-DER

$$\begin{aligned} p_1(k) &= \bar{p} \in [\underline{p}, \bar{p}], & p_2(k) &= \bar{p} \in [\underline{p}, \bar{p}] \\ e_1(k+1) &= 0.5(\bar{e} + \underline{e}) + \delta \cdot \bar{p} \in [\underline{e}, \bar{e}] \\ e_2(k+1) &= 0.5(\bar{e} + \underline{e}) + \delta \cdot \bar{p} \in [\underline{e}, \bar{e}]. \end{aligned}$$

Comparison of Examples 1 and 2 shows that certain feasible dispersions of $E(k)$ are preferable over others. The energy states $e_1(k)$ and $e_2(k)$ are a feasible dispersion of aggregated energy $E(k)$ both in Example 1 and in Example 2. However, the values for $e_1(k)$ and $e_2(k)$ chosen in Example 1 prevent the existence of a feasible dispersion of the desired (aggregated feasible) $E(k+1)$. The same $E(k+1)$ has instead a feasible dispersion with the values of $e_1(k)$ and $e_2(k)$ chosen in Example 2.

Motivated by these considerations, we denote

$$\mathcal{E}(k, E) \doteq \mathcal{E}(k) \cap s^{-1}(E) \quad (24)$$

as the set of feasible dispersions of E at time k .

Definition 1 (Consistent dispersion of $E(k)$): Given $E(k) \in \mathbb{R}$, the vector $\mathbf{e}(k) \in \mathbb{R}^N$ is said to be a *consistent dispersion* of $E(k)$ at time $k+1$, if $\mathbf{e}(k) \in \mathcal{E}(k, E(k))$ and

$$\mathcal{R}_{\text{agg}}(k+1) \equiv \mathcal{R}_{\text{agg}}^*(k+1)$$

with $\mathcal{R}_{\text{agg}}(k+1)$ as in (19) and $\mathcal{R}_{\text{agg}}^*(k+1)$ from (20). ■

The conceptual idea of consistent $\mathbf{e}(k)$ is illustrated in Fig. 5. Starting from the top-left corner, Fig. 5 shows that an aggregated state $E(k)$ can be dispersed in different ways, namely all the points in $\mathcal{E}(k, E(k))$. Among them, a consistent dispersion of the aggregated state $E(k)$ leads to a feasible and reachable set $\mathcal{R}(k+1)$ (on the right) whose propagation on the aggregated space, $\mathcal{R}_{\text{agg}}^*(k+1)$, is equivalent to the aggregated feasible and reachable set obtained via direct computation, i.e., from $E(k)$

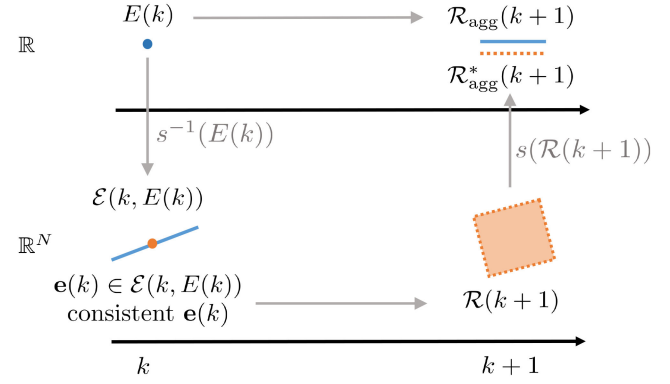


Fig. 5. Graphical representation of a consistent dispersion of $E(k)$.

and relative aggregated constraints as in (10). Considering the complete problem, the schedule $\{G(k)\}_{\mathcal{K}}$ can be computed via (9) as much as via (8) if $\mathbf{e}(k)$ is a consistent dispersion of $E(k)$ for all $k \in \mathcal{K}$, because the feasible set defined by (9b)–(9d) coincides with the one defined by (8b)–(8f) at each time step.

Remark 1 (Dispersions and two-stage scheduling): The concept of consistent dispersions helps understanding why (9) is successfully applied, cf., [12] and [15]. Therefore, hierarchical control is often used to solve (8) in a sequential manner. First, (9) is used [with (10)] on an upper level to provide long-term aggregated schedules. Second, lower level feedback control (e.g., MPC) assigns the dispersion of the aggregated energy state in a way that guarantees the feasibility of the precomputed schedule over the subsequent time steps. Essentially, this controller consistently disperses the aggregated schedule.

V. EXISTENCE AND COMPUTATION OF CONSISTENT DISPERSIONS

The considerations abovementioned motivate investigation of the existence of a consistent dispersion. To this end, we make the following assumption.

Assumption 2: For all $j \in \mathcal{N}$ and all $k \in \mathcal{K}$, let the sampling time δ and the intervals $\mathcal{P}_j(k)$, $\mathcal{E}_j(k)$, and $\mathcal{E}_j(k+1)$ be such that there exists at least one $e_j(k) \in \mathcal{E}_j(k)$ for which

$$(e_j(k) \oplus \delta \cdot \mathcal{P}_j(k)) \subseteq \mathcal{E}_j(k+1) \quad (25a)$$

$$(\mathcal{E}_j(k) \oplus \delta \cdot \mathcal{P}_j(k)) \supseteq \mathcal{E}_j(k+1) \quad (25b)$$

holds. ■

Condition (25a) implies that there exists a subset of $\mathcal{E}_j(k)$ from which all reachable states are feasible. The second part (25b) instead requires that the entire set $\mathcal{E}_j(k+1)$ is reachable from at least one $e_j(k) \in \mathcal{E}_j(k)$.

Moreover, consider $d_1 : \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}$ given by

$$\begin{aligned} d_1(k, \mathbf{e}(k)) &= \sum_{j \in \mathcal{N}} \left(e_j(k) + \delta \cdot \underline{p}_j(k) + \right. \\ &\quad \left. - \max \left\{ e_j(k) + \delta \cdot \underline{p}_j(k), \underline{e}_j(k+1) \right\} \right) \end{aligned} \quad (26a)$$

and $d_u : \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}$ given by

$$d_u(k, \mathbf{e}(k)) = \sum_{j \in \mathcal{N}} \left(e_j(k) + \delta \cdot \bar{p}_j(k) + \right. \\ \left. - \min \{ e_j(k) + \delta \cdot \bar{p}_j(k), \bar{e}_j(k+1) \} \right). \quad (26b)$$

Theorem 1 (Existence of consistent dispersions):

Suppose Assumptions 1 and 2 hold. If

$$\mathbf{e}(k) \in \arg \min_{\mathbf{e}(k) \in \mathcal{E}(k, E(k))} d_l(k, \mathbf{e}(k)) + d_u(k, \mathbf{e}(k)) \quad (27)$$

with d_l and d_u from (26), then $\mathcal{R}_{\text{agg}}(k+1)$ from (19) and $\mathcal{R}_{\text{agg}}^*(k+1)$ from (20) satisfy

$$\mathcal{R}_{\text{agg}}(k+1) \equiv \mathcal{R}_{\text{agg}}^*(k+1)$$

i.e., $\mathbf{e}(k)$ is a consistent dispersion of $E(k)$ at time $k+1$. ■

We give the proof in the appendix. Next, we briefly comment on the conditions required by Theorem 1. Assumptions 1 and 2 are quite mild, e.g., (25b) is trivially fulfilled when $\mathcal{E}(k)$ is constant (or decreasing with respect to set-inclusion) provided 0 belongs to $\mathcal{P}_j(k)$. However, they may require an additional modeling effort that is normally avoided. For instance, set relation (25b) requires exclusion of all the values of $\mathbf{e}(k+1)$, which are feasible from an energy perspective but that cannot be reached from any feasible $\mathbf{e}(k)$. The relation (16) can be satisfied by reducing the state constraints—i.e., if there are no feasible $e_j(k+1)$ reachable from $e_j(k)$, then this $e_j(k)$ should be excluded from $\mathcal{E}_j(k)$ —or by enlarging the set $\delta \cdot \mathcal{P}_j(k)$. Similarly, set relation (25a) can be imposed by reducing the set $\delta \cdot \mathcal{P}_j(k)$. To summarize, even if Assumptions 1 and 2 may not always hold for constraints modeling the exact EC-DER's physical limitations or operational requirements, they can be assumed without loss of generality after finding a suitable subset of the original constraint set. Finally, observe that Assumptions 1 and 2 are indirectly coupling the power and the energy constraints (2a) and (2b). In this respect, time-varying power constraints are useful to avoid that the most stringent energy constraint limits the power constraint over the entire scheduling horizon.

The cost function $d_l(k, \mathbf{e}(k)) + d_u(k, \mathbf{e}(k))$ mentioned in Theorem 1 involves the energy state $\mathbf{e}(k)$ *only* at time step k . This means that the computation of a consistent dispersion is independent from the knowledge of $\mathbf{e}(h)$ with $h \neq k$. Thus, $\mathbf{e}(k)$ (and consequently $\mathbf{p}(k-1)$) can be determined at each k by a lower level controller without any need for information about past or future states of the system. This has two important consequences: On the one hand, it avoids any dependency between dispersion at subsequent time instants in the scheduling problem, which complicates dealing with eventual random decision variables and parameters. On the other hand, the computation of $\mathbf{e}(k)$ does not require any load forecasts nor MPC strategies (the long-term perspective is already accounted for in the aggregated schedule). Note that while a consistent dispersion $\mathbf{e}(k)$ implies that the set of feasible dispersions $\mathcal{E}(k+1, E)$ is not empty, it does not guarantee that a consistent dispersion at $k+1$ exists. We address this point in the following theorem.

Theorem 2 (Recursively consistent dispersions): Suppose Assumptions 1 and 2 hold. If $E(k+1) \in \mathcal{E}_{\text{agg}}(k+1)$ and $\mathbf{e}(k)$

is a consistent dispersion of $E(k)$, then there exists at least one consistent dispersion $\mathbf{e}(k+1)$ of $E(k+1)$. ■

The proof is given in the appendix. An important consequence of Theorem 2 is that given that the initial $\mathbf{e}(0)$ is a consistent dispersion of $E(0)$, then a consistent dispersion of $\{P(k)\}_{\mathcal{K}}$ and $\{E(k+1)\}_{\mathcal{K}}$ is always possible, as long as (9c) and (9d) hold with (10). In other words, the aggregated schedule can always be tracked. Furthermore, note that while the existence of a consistent dispersion is a property of the entire system, the requirements of Assumptions 1 and 2 involve each EC-DER *separately*, which is in line with the task of aggregating heterogeneous devices. Finally, note that Theorem 2 induces an additional requirement for aggregated scheduling: $\mathbf{e}(0)$ must be a consistent dispersion of $E(0)$. Since $\mathbf{e}(0)$ is assigned, this requirement could theoretically compromise aggregated scheduling. However, whether $\mathbf{e}(k)$ is a consistent dispersion of $E(0)$ depends not only on the elements of $\mathbf{e}(k)$, but also on the constraints at k and $k+1$. Thus, a given $\mathbf{e}(0)$ can be enforced to be a consistent dispersion by replacing the power constraint set at $k=0$ and/or the energy constraint set at $k=1$ with an appropriate subset. We discuss this point with an example in Section VI-A.

Besides justifying the use of aggregated scheduling (9) in presence of a consistent dispersion of the aggregated energy state, Theorems 1 and 2 lead to further considerations in more general problems.

VI. DISCUSSION

Theorems 1 and 2 prove useful in at least two different contexts. First, they indicate a practical approach to tackle scheduling problems modeled via (3). Second, they open a path for theoretical studies on aggregated scheduling, where additional constraints extend (3). We provide a short example of both applications: Section VI-A discusses a practical case study, while Section VI-B deals with including conversion losses in the model.

A. Application Example

Theorems 1 and 2 indicate the following approach to the scheduling problem (3).

- 1) Check/impose consistency of constraints (3c)–(3e) with respect to Assumptions 1 and 2, and the requirement of $\mathbf{e}(0)$ being a consistent dispersion of $E(0)$.
- 2) Aggregate decision variables and constraint sets via the mapping s , as in (7) and (10).
- 3) Solve (9) to compute the schedule $\{G(k)\}_{\mathcal{K}}$.
- 4) Guarantee feasibility of the aggregated schedule by dispersing the aggregated variables following (27).

Among the practical applications for this approach there are the ones already discussed in the introduction. For the sake of completeness, we present, in this section, an additional example. Consider a population of N privately owned household batteries, whose dynamics are given by (1) and (2) (we discuss conversion losses in the following section and leave them out in this example for the sake of clarity). Each battery has its own controller, which

regulates the power output of the battery depending on the customized needs of its owner. However, the owners of the batteries also seek additional revenues by selling ancillary service to the utility as a community. Several companies are already providing such aggregating services, see for example [23] and [24]. We examine the case in which the population of batteries provides rise contingency reserves during times in which the batteries are not in use (e.g., at night). This reserve consists in a power injection that is required independently from the load; thus, we neglect the load and consider $G(k) = P(k)$. The population of batteries schedules a given level of aggregated power G_{ref} that it has to inject into the grid for a time interval of length t whenever the service is called.⁴ The population of batteries receives a payment proportional to $-G_{\text{ref}}$, even if the service is not called. Hence, the question to be answered is: what is the minimum G_{ref} that the population of batteries could confidently schedule at a given time of the day indicated with $k = 0$? Let us consider a discrete sampling interval $\delta = t = 0.25$ h, and a population of $N = 100$ domestic batteries with time-independent power limits, $\underline{p}_j = -4$ kW and $\bar{p}_j = 4$ kW for all $j \in \mathcal{N}$. With respect to the energy constraints, we assume time-varying constraints considering that the aggregated control must ensure a minimum energy state at the end of the time interval ($k = 1$), independently from e_j^0 . Let us take $\underline{e}_j(0) = 0$ kWh, $\underline{e}_j(1) = 2$ kWh, and $\bar{e}_j(0) = \bar{e}_j(1) = 13$ kWh for all $j \in \mathcal{N}$. Consider aggregation via projection s , as in (7) and (10). The minimum power that the population of batteries can collectively provide is $\underline{P} = -400$ kW. Does this value violate the aggregated energy constraints (9b) and (9d)? Imposing $P(0) = \underline{P} = -400$ kW, it follows $E(1) = E(0) - 100 \text{ kWh} \geq 200 \text{ kWh}$, which means that the choice of $P(0) = \underline{P}$ is feasible if $E(0) \geq 300$ kWh. The initial energy state $e(0)$ depends on previous operation, which is regulated by the local controller and unknown upon scheduling. Nonetheless, assume that it is possible to confidently forecast that $E(0) \geq 400$ kWh, e.g., by means of probabilistic forecasts on the aggregated energy [25]. From this reasoning, one could deduce that scheduling $G_{\text{ref}}(0) = -400$ kW is a sensible choice. However, the simulation tells otherwise.

We run a simulation assuming that each battery is constantly discharging at its maximum power $\underline{p}_j = -4$ kW. The simulation consists in implementing dynamics (1) with $\delta = 1$ min by means of a Python script.⁵ Like in real-world applications, we give priority to satisfying the constraints over tracking the schedule. Thus, we force an increase of p_j whenever required such that $e_j(1)$ satisfies the energy constraints. Fig. 6 depicts the trajectories of P and E . The initial energy state of each battery is such that $E(0) \geq 400$ kWh. However, while $E(0)$ satisfies the requirement of being higher than 300 kWh, P diverges from the reference (depicted in Fig. 6 with a dot-dash line). The reason can be deduced from the trajectories of e_j and p_j plotted in Fig. 7 for all $j \in \mathcal{N}$. Therefore, the dash lines highlight the power and energy limits. As can be seen, some batteries start

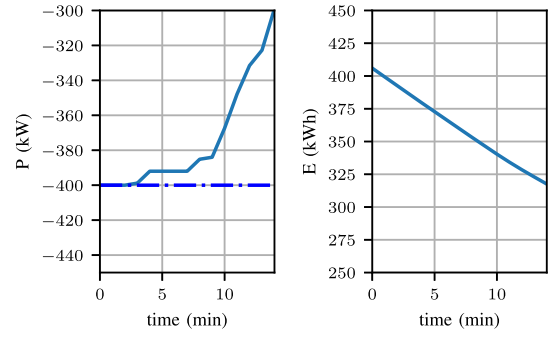


Fig. 6. Aggregated power and energy with $e(0)$ not being a consistent dispersion of $E(0)$.

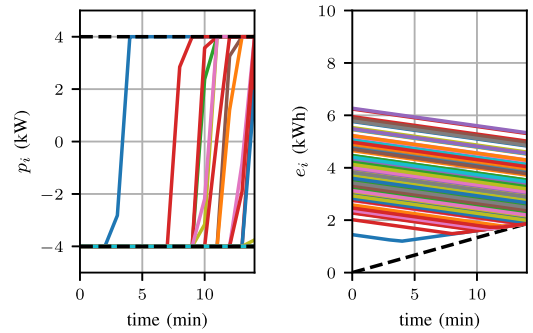


Fig. 7. Power and energy of each battery with $e(0)$ not being a consistent dispersion of $E(0)$.

at a low energy state, or get discharged over time until a point from which they must invert the direction of their power output (charging) to avoid a violation of the energy constraint. Thus, even if $P(0)$ and $E(1)$ would satisfy the aggregated constraints, the aggregated schedule cannot be tracked. In view of Theorems 1 and 2, this phenomenon is to be expected.

The proposed test case meets both Assumptions 1 and 2. However, the given $e(0)$ is not a consistent dispersion of $E(0)$. Thus, it is not guaranteed that an aggregated feasible schedule can be tracked in practice. Besides explaining the failure of the aggregated approach, Theorems 1 and 2 suggest a possible solution. The energy states $e(0)$ are given and cannot change. Nonetheless, as already commented in Section V, an appropriate modification of the constraints set can render $e(0)$ a consistent dispersion of $E(0)$. It is easily verified that if the lower power limit is modified as

$$\underline{p}_j(e_j(0)) \doteq \begin{cases} \underline{p}_j = -4 \text{ kW} & \text{if } e_j(0) \geq 3 \text{ kWh} \\ \underline{p}_j = \frac{(e_j(0) - \underline{e}_j(1))}{t} & \text{if } e_j(0) < 3 \text{ kWh} \end{cases}$$

then any $e(0)$ is a consistent dispersion of $E(0)$.⁶

Returning to the scheduling problem, we have now that $\underline{p}_j$ depends on $e_j(0)$, unknown upon bidding. Even in this case, one could create probabilistic forecasts for the lower limit of the aggregated power output $\underline{P}$, by propagating the distribution

⁴Recall that in the notation used in this article a grid power injection corresponds to a negative G .

⁵Note that the sampling interval of the simulation differs from the sampling interval of the scheduling. This allows for a better observation of the occurring phenomena.

⁶Note that $\underline{p}_j(e_j(0))$ cannot be greater than $\bar{p}_j$. If $\underline{p}_j(e_j(0)) \geq \bar{p}_j$ for any $j \in \mathcal{N}$, then it is not possible to obtain a consistent dispersion by changing the feasible set of p_j . However, this case correspond to an ill-defined constraint set and does not constitute a severe limitation of the proposed method.

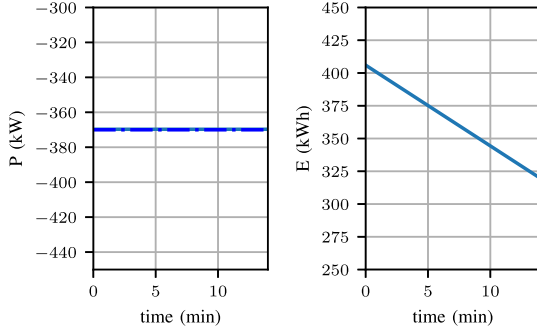


Fig. 8. Aggregated power and energy with $e(0)$ being a consistent dispersion of $E(0)$.

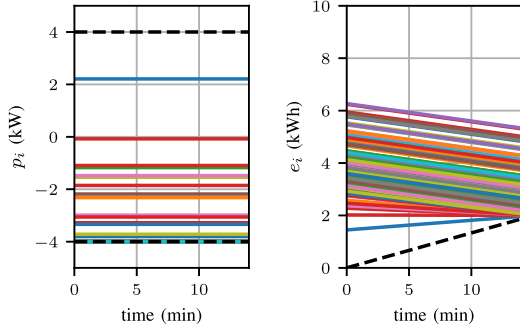


Fig. 9. Power and energy of each battery with $e(0)$ being a consistent dispersion of $E(0)$.

of $e_j(0)$ to $\underline{p}_j(e_j(0))$. Assume that such forecasts indicate that the probability of having $\underline{P} \geq -370$ kW is $\simeq 1$; then bidding $G_{\text{ref}} = -370$ kW is a reasonable choice.

A new problem arises: how should the aggregated schedule be dispersed among the N batteries? The actual $\underline{P}$, computed once $e(0)$ is known, may be lower than the chosen $G_{\text{ref}} = -370$ kW. Therefore, operating each battery at its maximum $\underline{p}_j(e_j(0))$ does not guarantee tracking of G_{ref} . At the same time, requiring all batteries to provide $p_j(0) = G_{\text{ref}}/N$ does not work either, similar to the previous case (some of the batteries get discharged before others and stop providing the required power output prior to $k = 1$). Indeed, the concept of consistent dispersion allows to easily tackle this problem: i) given $e(0)$ (and, therefore, $\underline{p}_j(e_j(0))$ for all $j \in \mathcal{N}$), calculate a consistent dispersion $e(1)$ of $E(1) = s(e(0)) + \delta \cdot G_{\text{ref}} = s(e(0)) - 92,5$ kWh; ii) apply $\mathbf{p}(k) = \delta^{-1}(e(1) - e(0))$. We simulate this latter control as for the previously discussed case. From Fig. 8, it can be seen that the chosen G_{ref} is met over the entire time interval without violating any constraint. Fig. 9 illustrates the details of each battery.

B. Conversion Losses

The method discussed in the previous section is applicable only to problems that can be modeled by (3), mostly market-based problems hinging around plain power balance. Limitations of (3) are: a) it neglects nonlinearities in the EC-DERs dynamics (3c); b) it ignores constraints that involve a partial summation of the elements of $\mathbf{p}(k)$ and $\mathbf{e}(k)$; c) it overlooks the flow of reactive power. In power systems, nonlinearities in

the EC-DERs dynamics are used to model conversion losses, while network constraints comprise partial summations and requires considering the flow of reactive power. Hence, additional research work is needed to extend applicability of the proposed approach. In the following, we discuss Issue a), while Issues b) and c) are left out due to space limitations and will be addressed in future works.

B. Modeling Conversion Losses: We seek a constraint reformulation that allows using Theorems 1 and 2 for an aggregation of nonlinear dynamics. To this end, consider a dynamic model of the energy state including conversion losses, for example

$$e_j(k+1) = e_j(k) + \delta \cdot p_j(k) - \mu_j(p_j(k))\delta \cdot |p_j(k)| \quad (28)$$

where the energy lost in conversion is $\mu_j(p_j(k))\delta \cdot |p_j(k)|$. Note that $\mu_j(p_j(k))$ can be a function of $p_j(k)$, see [26]. The dynamics (28) contain an energy-based description of the storage losses. This model is often applied in energy management even if it simplifies the phenomena occurring in practice; nonetheless, it has been established experimentally to be sufficiently accurate for scheduling purposes, cf., [27].

Next, consider an alternative—yet equivalent—model describing the conversion losses in terms of lost power. First, we define

$$\tilde{p}_j(k) = p_j(k) - \mu_j(p_j(k)) \cdot |p_j(k)| \quad (29a)$$

as the actual power exchange with the storage. Then, we substitute $\tilde{p}_j(k)$ into (28), which yields

$$e_j(k+1) = e_j(k) + \delta \cdot \tilde{p}_j(k). \quad (29b)$$

Given (29a), the power constraint (2a) can be equivalently expressed as a constraint on $\tilde{p}_j(k)$, i.e.,

$$\tilde{\mathcal{P}}(k) = \{\tilde{\mathbf{p}}(k) \mid \exists \mathbf{p}(k) \in \mathcal{P}(k) \text{ s.t. (29a) holds}\}. \quad (29c)$$

In other words, conversion losses modify the boundaries of the set of reachable states at $k+1$. Note that $\tilde{\mathcal{P}}(k)$ is a closed real interval.

Finally, consider the scheduling problem (8) including conversion losses as modeled by (29a)–(29c)

$$\min_{\substack{\{G(k)\}_{k \in \mathcal{K}}, \\ \{\tilde{\mathcal{P}}(k)\}_{k \in \mathcal{K}}, \{E(k+1)\}_{k \in \mathcal{K}}, \\ \{\tilde{\mathbf{p}}(k)\}_{k \in \mathcal{K}}, \{e(k+1)\}_{k \in \mathcal{K}}}} \sum_{k \in \mathcal{K}} c(G(k)) \quad (30a)$$

$$\text{s. t. } \forall k \in \mathcal{K}$$

$$G(k) = \tilde{\mathcal{P}}(k) - L(k) - P_{\text{loss}}(k) \quad (30b)$$

$$e(k+1) = e(k) + \delta \cdot \tilde{\mathbf{p}}(k) \quad e(0) = e^0 \quad (30c)$$

$$\tilde{\mathbf{p}}(k) \in \tilde{\mathcal{P}}(k) \quad (30d)$$

$$e(k+1) \in \mathcal{E}(k+1) \quad (30e)$$

$$\tilde{\mathcal{P}}(k) = s(\tilde{\mathbf{p}}(k)) \quad (30f)$$

$$E(k+1) = s(e(k+1)) \quad (30g)$$

$$P_{\text{loss}}(k) = f(\tilde{\mathbf{p}}(k)). \quad (30h)$$

The function $f(\tilde{\mathbf{p}}(k))$ returns the total conversion losses (in terms of power) resulting from $\tilde{\mathbf{p}}(k)$.⁷ Equation (30h) is the only practical difference between problems (30) and (8). Thus, the question arises if the solution of (30) can be approached in two steps (aggregated scheduling and subsequent dispersion), as for (8), without compromising the result.

The aggregated version of (30) reads

$$\min_{\substack{\{G(k)\}_{\mathcal{K}}, \\ \{\tilde{P}(k)\}_{\mathcal{K}}, \{E(k+1)\}_{\mathcal{K}}}} \sum_{k \in \mathcal{K}} c(G(k)) \quad (31a)$$

$$\text{s. t. } \forall k \in \mathcal{K}$$

$$G(k) = \tilde{P}(k) - L(k) - f_{\text{agg}}(\tilde{P}(k)) \quad (31b)$$

$$E(k+1) = E(k) + \delta \cdot \tilde{P}(k), \quad E(0) = s(e^0) \quad (31c)$$

$$\tilde{P}(k) \in \tilde{\mathcal{P}}_{\text{agg}}(k) \quad (31d)$$

$$E(k+1) \in \mathcal{E}_{\text{agg}}(k+1). \quad (31e)$$

Observe that constraints (30c)–(30g) and (31c)–(31e) have the exact same structure of the corresponding ones in (8) and (9). Therefore, the idea of a consistent dispersion and the statements of Theorems 1 and 2 still hold. The main difficulty is represented here by enforcement of the equivalence between (31b) and (30b), (30h).

B. Aggregated Losses: Consider a homogeneous population where the losses are the same on each device and independent of $p_j(k)$, i.e., $\mu_j(p_j(k)) = \mu$ for all $j \in \mathcal{N}$. Furthermore, consider the absence of mutual exchange of power among devices, i.e., $\text{sign}(p_j(k)) = \text{sign}(p_i(k))$ for all $\{j, i\} \in \mathcal{N} \times \mathcal{N}$. Then, defining

$$f_{\text{agg}}(\tilde{P}(k)) = \begin{cases} \frac{\mu}{1-\mu} & \text{if } \tilde{P}(k) \geq 0 \\ \frac{-\mu}{1+\mu} & \text{if } \tilde{P}(k) < 0 \end{cases} \quad (32)$$

the equivalence $f_{\text{agg}}(\tilde{P}(k)) = f(\tilde{\mathbf{p}}(k))$ holds exactly for each $\tilde{P}(k) \in \tilde{\mathcal{P}}_{\text{agg}}(k)$. Thus, in this specific case, (31b) is equivalent to (30b), (30h), and aggregation as in (31) leads to the same solution of (30). However, in general $f_{\text{agg}}(\tilde{P}(k))$ can only approximate $P_{\text{loss}}(k)$ and the solution of (31) deviates from the one of the original problem (30) depending on the severity of this approximation. This aspect might compromise the effectiveness of aggregation. Eventually, aggregation of groups of EC-DERs with similar conversion losses can reduce the severity of the approximation at the price of a slightly higher number of variables in the aggregated problem.

VII. CONCLUSION

This article investigated scheduling for energy systems based on an aggregated time-varying model of heterogeneous energy-constrained distributed energy resources. The aggregated model reduces the number of variables in the scheduling problem—thus lowering the computational burden and simplifying the management of the uncertainties. However, aggregated models

are not trivial to compute. The methods employed in the literature either require to predetermine the dispersion rule, or entail a relaxation that may enlarge the feasible set to values that cannot be attained in practice. This article took a slightly different approach: we analyzed the nature of the aggregated constraints to avoid problems of enlarged feasible sets on the aggregated level. We show that, whenever the energy states of the individual devices satisfy a specific (collective) property—existence of a *consistent dispersion of the aggregated energy state* at the first time-step—the relaxed aggregation does not alter the feasible set and, thus, does not imply any optimality loss. Our main contribution is to prove that a consistent dispersion always exists under mild assumptions on the constraints of the various devices, considered separately. Our findings justify the application of an aggregated model in scheduling. Moreover, they provide a technique to check the consistency of an aggregated model. They allow proposing a way to disperse the aggregated energy state at each time separately, which is guaranteed to be recursively feasible. Furthermore, our results allow analyzing advantages and limitations of aggregated models in scheduling. Future work will apply the described methods to a real test case and extend the analysis to include network constraints with consideration of reactive power flow.

APPENDIX PROOF OF THEOREMS 1 AND 2

Proof of Theorem 2

The main idea behind the proof of Theorem 1 is showing that the bounds of the real intervals $\mathcal{R}_{\text{agg}}(k+1)$ and $\mathcal{R}_{\text{agg}}^*(k+1)$ coincide when $\mathbf{e}(k)$ satisfies (27). First, we prove two technical lemmata to then turn towards the proof of Theorem 1.

From (19), it follows that there are two possible cases for each of the two bounds of $\mathcal{R}_{\text{agg}}(k+1)$.

(1) Case (i):

$$E(k) + \delta \cdot \underline{P}(k) \geq \underline{E}(k+1) \quad (33a)$$

$$E(k) + \delta \cdot \bar{P}(k) \leq \bar{E}(k+1). \quad (33b)$$

(2) Case (ii):

$$E(k) + \delta \cdot \bar{P}(k) > \bar{E}(k+1) \quad (34a)$$

$$E(k) + \delta \cdot \underline{P}(k) < \underline{E}(k+1). \quad (34b)$$

Observe that, differently from Case (i), the inequalities in Case (ii) are strict. Furthermore, note that the feasible and reachable states at time k are limited by the reachability (power) constraint (9c) in Case (i), and by the feasibility (energy) constraint in Case (ii). We analyze both cases in two technical lemmata.

Lemma 1 (Case (i)): Suppose Assumptions 1 and 2 hold and let $\mathbf{e}(k)$ satisfy (27).

1) If (33a) holds, then the lower bound $\underline{E}^r(k+1)$ of $\mathcal{R}_{\text{agg}}(k+1)$ from (19) and the lower bound $\sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1)$ of $\mathcal{R}_{\text{agg}}^*(k+1)$ from (20) satisfy

$$\underline{E}^r(k+1) = \sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1).$$

⁷Note that depending on the relation between $\mu_j(p_j(k))$ and $p_j(k)$, $f_{\text{loss}}(\tilde{\mathbf{p}}(k))$ may not even have a closed form description.

- 2) If (33b) holds, then the upper bound $\bar{E}^r(k+1)$ of $\mathcal{R}_{\text{agg}}(k+1)$ from (19) and the upper bound $\sum_{j \in \mathcal{N}} \bar{e}_j^r(k+1)$ of $\mathcal{R}_{\text{agg}}^*(k+1)$ from (20) satisfy

$$\bar{E}^r(k+1) = \sum_{j \in \mathcal{N}} \bar{e}_j^r(k+1).$$

Proof: First, consider the lower bound case (a), i.e., that (33a) holds. From $E(k) + \delta \cdot \underline{P}(k) \geq \underline{E}(k+1)$, we have

$$E(k) \geq -\delta \cdot \underline{P}(k) + \underline{E}(k+1).$$

Thus, for a $E(k) \in \mathcal{E}_{\text{agg}}(k) = s(\mathcal{E}(k))$, there exists a $\mathbf{e}(k) = s^{-1}(E(k))$ such that

$$e_j(k) \geq -\delta \cdot \underline{p}_j(k) + \underline{e}_j(k+1) \quad \forall j \in \mathcal{N}. \quad (35)$$

Equation (25a) in Assumption 2 implies that there is at least one $\mathbf{e}(k)$ satisfying (3c)–(3e) for which

$$e_j(k) - \delta \cdot \underline{p}_j(k) \geq \underline{e}_j(k+1) \quad \forall j \in \mathcal{N}.$$

Thus, there exists at least one feasible $\mathbf{e}(k) = s^{-1}(E(k))$ for which (35) holds. In other words, there exists a $\mathbf{e}(k) \in \mathcal{E}(k, E(k))$ satisfying (35).

Observe that $d_1(k, \mathbf{e}(k))$ reaches its global minimum zero for any $\mathbf{e}(k)$ satisfying (35). Furthermore, from (25a) in Assumption 2, it holds that

$$\bar{e}_j(k+1) - \delta \cdot \bar{p}_j(k) \geq \underline{e}_j(k+1) - \delta \cdot \underline{p}_j(k). \quad (36)$$

This implies that diminishing the value of any $e_j(k)$ below $-\delta \cdot \underline{p}_j(k) + \underline{e}_j(k+1)$ does not decrease $d_u(\mathbf{e}(k), k)$. Therefore, (27) for Case (i)–(a) implies (35).

In turn, it follows from (35) that:

$$e_j(k) + \delta \cdot \underline{p}_j(k) \geq \underline{e}_j(k+1) \quad \forall j \in \mathcal{N}.$$

Therefore, the lower bound of $\mathcal{R}_{\text{agg}}^*(k+1)$ is equal to

$$\sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1) = \sum_{j \in \mathcal{N}} e_j(k) + \delta \cdot \sum_{j \in \mathcal{N}} \underline{p}_j(k) = \underline{E}^r(k+1).$$

The second part of Case (i), $E(k) + \delta \cdot \bar{P}(k) \leq \bar{E}(k+1)$, follows mutatis mutandis. It is skipped for the sake of brevity. ■

Lemma 2 (Case (ii)): Suppose Assumptions 1 and 2 hold and let $\mathbf{e}(k)$ satisfy (27).

- 1) If (34a) holds, then the lower bound $\underline{E}^r(k+1)$ of $\mathcal{R}_{\text{agg}}(k+1)$ from (19) and the lower bound $\sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1)$ of $\mathcal{R}_{\text{agg}}^*(k+1)$ from (20) satisfy

$$\underline{E}^r(k+1) = \sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1).$$

- 2) If (34b) holds, then the upper bound $\bar{E}^r(k+1)$ of $\mathcal{R}_{\text{agg}}(k+1)$ from (19) and the upper bound $\sum_{j \in \mathcal{N}} \bar{e}_j^r(k+1)$ of $\mathcal{R}_{\text{agg}}^*(k+1)$ from (20) satisfy

$$\bar{E}^r(k+1) = \sum_{j \in \mathcal{N}} \bar{e}_j^r(k+1).$$

Proof (By contradiction): First, consider the lower bound case (b), i.e., that (34a) holds. Assume for the sake of contradiction that there is a group of devices $\mathcal{I} \subset \mathcal{N}$ for which

$$e_i(k) + \delta \cdot \underline{p}_i(k) > \underline{e}_i(k+1) \quad \forall i \in \mathcal{I} \quad (37)$$

and another group of devices $\mathcal{H} = \mathcal{N} \setminus \mathcal{I}$ for which

$$e_h(k) + \delta \cdot \underline{p}_h(k) \leq \underline{e}_h(k+1) \quad \forall h \in \mathcal{H}. \quad (38)$$

From (7), it follows that $E(k) = \sum_{\mathcal{I}} e_i(k) + \sum_{\mathcal{H}} e_h(k)$.

Suppose that $\mathcal{H} = \emptyset$, then (34a) is false and Case (i) applies. Hence, $\mathcal{H} \neq \emptyset$. Moreover, for at least one $\tilde{h} \in \mathcal{H}$, Inequality (38) holds strictly (otherwise again Case (i) applies).

Consider a small positive increment $\Delta > 0$ for which $\tilde{e}_{\tilde{h}}(k) = e_{\tilde{h}}(k) + \Delta$ still satisfies (38). The increment of $\tilde{e}_{\tilde{h}}(k)$ is always allowed. In fact, given Assumption 2, we have $\tilde{e}_{\tilde{h}}(k) + \delta \cdot \underline{p}_{\tilde{h}}(k) \geq \underline{e}_{\tilde{h}}(k+1)$. Thus, from (38) (which is strictly satisfied for $\tilde{h}$), it follows that $e_{\tilde{h}}(k) < \bar{e}_{\tilde{h}}(k)$.

Next, consider w.l.o.g. one EC-DER $\tilde{i} \in \mathcal{I}$ for which the energy $\tilde{e}_{\tilde{i}}(k) = e_{\tilde{i}}(k) - \Delta$ is decreased such that $\tilde{e}_{\tilde{i}}(k)$ still satisfies (37). This reduction is always possible: given Assumption 2, we have $\underline{e}_{\tilde{i}}(k) + \delta \cdot \underline{p}_{\tilde{i}}(k) \leq \underline{e}_{\tilde{i}}(k+1)$. Thus, (37) implies $e_{\tilde{i}}(k) > \underline{e}_{\tilde{i}}(k)$.

Increasing the energy at EC-DER $\tilde{h}$ and reducing it at the same time at EC-DER $\tilde{i}$ satisfies (7), as

$$E(k) = -\Delta + \sum_{\mathcal{I}} e_i(k) + \sum_{\mathcal{H}} e_h(k) + \Delta.$$

However, $d_1(\tilde{\mathbf{e}}(k), k) < d_1(\mathbf{e}(k), k)$. Moreover, $d_u(\tilde{\mathbf{e}}(k), k) \leq d_u(\mathbf{e}(k), k)$. In fact, the increment of $e_{\tilde{h}}(k)$ cannot increase $d_u(\mathbf{e}(k), k)$, because the contribution of the components $e_h(k)$ to $d_u(\mathbf{e}(k), k)$ is zero.⁸ Thus, $\mathbf{e}(k)$ does not minimize $d_1(\mathbf{e}(k), k) + d_u(\mathbf{e}(k), k)$. We arrive at a contradiction, i.e., $\mathcal{I} \neq \emptyset$ contradicts (27). Hence, we arrive at $\mathcal{I} = \emptyset$.

Given that (38) hold for all the EC-DERs, then the lower bound of $\mathcal{R}_{\text{agg}}^*(k+1)$ is equal to

$$\sum_{j \in \mathcal{N}} \underline{e}_j^r(k+1) = \sum_{j \in \mathcal{N}} \underline{e}_j(k+1) = \underline{E}^r(k+1).$$

Summing up, the lower bounds of $\mathcal{R}_{\text{agg}}(k+1)$ and $\mathcal{R}_{\text{agg}}^*(k+1)$ also match in Case (ii)–(a) under (27). In a similar fashion, it can be shown that the upper bound of $\mathcal{R}_{\text{agg}}(k+1)$ equals the upper bound of $\mathcal{R}_{\text{agg}}^*(k+1)$ in Case (ii)–(b) under (27). ■

Now, we can finalize the proof of Theorem 1. As we have seen in Lemma 1 for Case (i) and in Lemma 2 for Case (ii), if (27) holds then the upper and lower bound of $\mathcal{R}_{\text{agg}}(k+1)$ and $\mathcal{R}_{\text{agg}}^*(k+1)$ coincide. Hence, we have that $\mathcal{R}_{\text{agg}}(k+1) \equiv \mathcal{R}_{\text{agg}}^*(k+1)$. □

Proof of Theorem 2

By construction, if $\mathbf{e}(k)$ is a consistent dispersion of $E(k)$ and $E(k+1) \in \mathcal{E}_{\text{agg}}(k+1)$, then there exist at least a feasible dispersion of $E(k+1)$, cf., Definition 1. Thus, the set $\mathcal{E}(k+1, E(k+1))$ is not empty. The cost function $d_1(k+1$

⁸Given (36) then $\tilde{e}_{\tilde{h}}(k) \leq \bar{e}_{\tilde{h}}(k+1) - \delta \cdot \bar{p}_{\tilde{h}}(k)$.

$1, \mathbf{e}(k+1)) + d_u(k+1, \mathbf{e}(k+1))$ is real-valued and continuous and the set $\mathcal{E}(k+1, E(k+1))$ is compact. Thus, by virtue of the extreme value theorem, the function $d_1(k+1, \mathbf{e}(k+1)) + d_u(k+1, \mathbf{e}(k+1))$ attains a minimum in $\mathcal{E}(k+1, E(k+1))$, i.e., at least one minimizer exists in (27). Consequently, applying Theorem 1, there exist a consistent dispersion of $E(k+1)$. ■

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